

Customer Purchase Pattern Analyzer

Executive Business Intelligence & Customer Analytics Report • Retail E-Commerce

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Technology Stack: Python 3.12, Pandas, NumPy, Scikit-Learn, SQLite, Plotly, Streamlit, Pytest, ReportLab

GitHub Repository: customer-purchase-pattern-analyzer

1. Executive Summary

This analytics engineering and customer intelligence project provides end-to-end transaction analysis, RFM cohort segmentation, and customer lifetime value optimization for an omnichannel Indian retail enterprise.

Analyzing **1,160 validated transactions** across **129 distinct customer profiles** over calendar year 2024, the pipeline generated **Rs. 3,771,206.53** in total gross merchandise volume with an Average Order Value of **Rs. 3,251.04**.

Key strategic findings reveal that **88.37% of customers are repeat buyers** generating **99.01% of company revenue**, customer lifetime equity yields a **13.1x spend multiplier** upon second-order conversion, and the top **40.3% of customers generate 80.0% of all gross sales**. Furthermore, **32 high-value repeat spenders** are currently overdue for purchase based on their historical purchase cadence, presenting a quantifiable **Rs. 476,000+ retention intervention opportunity**.

2. Business Problem & Strategic Objectives

Modern retail and e-commerce organizations struggle with disjointed transactional data, leading to three critical strategic vulnerabilities:

- **Undifferentiated Mass Marketing:** Treating high-value repeat spenders and one-time discount seekers identically, causing customer fatigue and wasted promotional spend.
- **Silent Customer Churn:** Inability to detect when a loyal buyer has exceeded their natural replenishment cadence until after they have permanently defected.
- **Margin Dilution:** Promoting low-margin hardware categories (e.g. electronics at 25.4% gross margin) without enforcing cross-sell attach rates for high-margin categories (e.g. beauty at 62.8% margin).

Strategic Project Goals:

- Build an automated, reproducible data engineering pipeline that cleans raw transactional feeds, imputes missing values with justified domain rules, and detects outliers using category-specific IQR fences.
- Engineer multi-dimensional behavioral features including calendar seasonality, demographic groupings, customer tenure, purchase frequency, recency, and 3-year Basic Customer Lifetime Value (CLV).
- Construct rule-based RFM (Recency, Frequency, Monetary) quintile cohorts and compare them against unsupervised K-Means clustering (k=4).
- Deliver an interactive executive dashboard with corporate styling and automated dynamic business recommendations.

3. Dataset Architecture & Preprocessing

3.1 Data Schema

The raw transactional log (customer_purchases_raw.csv) comprises 1,235 raw records across 20 attributes:

- **Demographics:** Customer_ID, Customer_Name, Age, Gender, City, Region, Occupation
- **Transactions:** Product_Category, Product_Name, Purchase_Date, Quantity_Purchased, Unit_Price, Discount_Used, Total_Purchase_Value
- **Operational:** Payment_Method, Purchase_Channel, Loyalty_Status, Customer_Segment

3.2 Cleaning & Preprocessing Summary (clean_data.py)

The automated data cleaning suite enforced 8 rigorous validation steps:

- **Duplicate Removal:** Removed 35 exact duplicate rows (1,235 → 1,200 rows).
- **Name Standardization:** Standardized 949 names (trimmed whitespace, collapsed spaces, formatted to Title Case).
- **Date Normalization:** Parsed mixed date formats into ISO YYYY-MM-DD; flagged and dropped 22 unparseable dates.
- **Taxonomy Normalization:** Corrected 97 category typos and casing mismatches using fuzzy string matching (difflib).
- **Missing Value Imputation:** Dropped 40 records missing critical keys (Customer_ID or Purchase_Date). Recomputed 24 missing totals mathematically (Qty \times Price \times (1 - Discount)). Imputed Age via median (47.0 years) and categorical attributes via statistical mode.
- **Outlier Detection (IQR Method):** Evaluated category-specific fences ([Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]). Identified 82 outliers and **flagged** them (Is_Outlier = True) without deletion to preserve financial accounting reconciliation.
- **Validation Test Suite:** 100% pass rate across zero-quantity, temporal bounds, and formula reconciliation tests. Final clean dataset: **1,160 rows, 21 columns**.

4. Analytical Metric Definitions & Mathematical Formulations

Metric Name	Mathematical Formula	Business Purpose
Total Revenue	$\text{Revenue} = \sum_{i=1}^N \text{Total_Purchase_Value}_i$	Measure gross merchandise volume
Average Order Value (AOV)	$\text{AOV} = \frac{\sum \text{Total_Purchase_Value}}{\text{Total_Orders}}$	Measure transaction basket magnitude
Purchase Frequency	$\text{Frequency} = \frac{\text{Total_Orders}}{\text{Distinct_Customers}}$	Measure customer velocity per customer
Recency	$\text{Recency} = \max(\text{Date}) - \text{Last_Purchase_Date}$	Measure recency of customer engagement (days)
Repeat Purchase Rate	$\text{Repeat Rate} = \frac{\text{Count}(\text{Customers with } \geq 2 \text{ orders})}{\text{Total_Customers}} \times 100$	Measure customer loyalty and repeat business
Gross Margin %	$\text{Margin \%} = \frac{\text{Total Revenue} - \text{Total COGS}}{\text{Total Revenue}} \times 100$	Measure profitability and pricing efficiency
Basic CLV	$\text{CLV} = \text{AOV} \times \text{Frequency} \times \frac{1}{\text{Discount Rate}}$	Measure customer lifetime equity
Market Basket Lift	$\text{Lift}(A, B) = \frac{P(A \cap B)}{P(A) \times P(B)}$	Measure of co-purchase affinity above chance

5. Customer Segmentation Methodology

5.1 Rule-Based RFM Quintile Scoring

Customers were scored from 1 to 5 across Recency (lower recency = higher score 5), Frequency (higher orders = 5), and Monetary value (higher spend = 5). Customers were mapped into 7 actionable cohorts:

- **Champions (32 customers, 24.8% base | Rs. 2,118,754.10, 56.2% rev):** Bought recently, buy frequently, spend highest.
- **Loyal Customers (26 customers, 20.2% base | Rs. 804,174.56, 21.3% rev):** Steady purchasing cadence and strong profitability.
- **Can't Lose Them (11 customers, 8.5% base | Rs. 364,776.46, 9.7% rev):** Historically heavy spenders who have not purchased in >70 days.
- **At Risk (19 customers, 14.7% base | Rs. 196,343.04, 5.2% rev):** Below-average recency and declining frequency.
- **Hibernating (31 customers, 24.0% base | Rs. 142,976.09, 3.8% rev):** Lapsed buyers with low frequency (avg recency: 192 days).
- **New Customers (5 customers, 3.9% base | Rs. 85,627.33, 2.3% rev):** High recency (bought <10 days ago), low order count.
- **Potential Loyalists (5 customers, 3.9% base | Rs. 58,554.95, 1.6% rev):** Recent buyers with average frequency.

5.2 Unsupervised K-Means Clustering (k=4)

Using log-transformed and standardized RFM features, the optimal cluster count k=4 was selected via Elbow Method and Silhouette Analysis (Silhouette Score: **0.4313**):

- **Cluster 0 — Core VIP Champions (42 customers):** Avg spend Rs. 64,288, avg frequency 15.6 orders, recency 14 days.
- **Cluster 1 — Lapsed / Low-Frequency Buyers (36 customers):** Avg spend Rs. 4,892, avg frequency 2.1 orders, recency 188 days.
- **Cluster 2 — Steady Mid-Tier Spenders (28 customers):** Avg spend Rs. 18,450, avg frequency 7.8 orders, recency 32 days.
- **Cluster 3 — Occasional / Developing Buyers (23 customers):** Avg spend Rs. 9,210, avg frequency 4.5 orders, recency 58 days.

6. Key Business Findings & Strategic Insights

- **Severe Revenue Concentration:** Pareto analysis proves that the top **40.3% of customers generate 80.0% of revenue**. Top 20% of customers generate 57.3% of sales.
- **Second-Purchase Lifetime Multiplier:** Repeat buyers spend **Rs. 32,752.20 vs Rs. 2,497.05 for one-time buyers** — a **13.1x lifetime spend multiplier**.
- **Category Profitability Skew:** Beauty (62.8% margin) and Books (57.2% margin) deliver superior gross margin percentages, whereas Electronics (25.4% margin) commands scale but dilutes gross margin.
- **Loyalty Status Expansion:** Platinum members generate a **Rs. 4,175.23 AOV** (+34.3% higher than Regular members), validating tiered loyalty incentives.
- **Geographic Balance:** North leads total revenue (Rs. 1,068,592.56, 28.3% share), while East commands the highest average order value (Rs. 3,860.22).
- **Seasonal Demand Spikes:** Twin annual peaks in April (Rs. 476,871.00, +99.8% MoM) and December (Rs. 336,084.00, +58.5% MoM).
- **Cross-Category Affinity:** Strong association lift between complementary items (e.g. Electric Blender + Face Cleanser, 2.39x Lift; Blender + Power Bank, 2.24x Lift).
- **Quantifiable Churn Risk:** 32 repeat-capable customers are overdue for purchase based on personal inter-purchase gaps, and 61 customers across churn-risk RFM cohorts represent Rs. 704,000+ in historical spend.

7. Actionable Strategic Recommendations

- **Implement VIP Retention Concierge:** Protect the 32 Champions and 11 "Can't Lose Them" accounts with priority support, zero-fee expedited shipping, and personalized quarterly gifts.
 - **Automate Second-Order Bridge Nurture:** Trigger automated SMS/email flows 14 days post-first purchase offering a 10% voucher for high-margin categories.
 - **Enforce Cross-Sell Bundling at Checkout:** Configure automated cross-sell carousels pairing low-margin Electronics with high-margin Beauty or Accessories to maintain blended margins >40%.
 - **Execute Win-Back Campaigns for Overdue Customers:** Deploy targeted 15% discount incentives with time limits for the 32 customers identified in the overdue retention audit.
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8. Limitations & Assumptions

- **Synthetic Dataset:** The dataset is synthetically generated to mirror real-world e-commerce dynamics; findings reflect the underlying synthetic distributions.
 - **Unit Cost & Margin Assumption:** Unit Cost is estimated via industry-standard COGS heuristics (Electronics: 70%, Beauty: 35%, Clothing: 45%, Books: 40%). Actual manufacturer rebates, shipping overhead, and return rates are omitted.
 - **Fixed CLV Lifespan:** Basic CLV assumes a fixed 3.0-year horizon ($AOV \times \text{Frequency} \times 3.0$). In production, probabilistic models should dynamically estimate individual churn probability.
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9. Future Analytics Roadmap

- **Machine Learning Churn Prediction:** Train XGBoost and Random Forest binary classifiers on customer feature vectors to predict 90-day churn probability with SHAP interpretability.
- **Personalized Collaborative Filtering:** Build an item-item collaborative filtering recommendation engine using implicit matrix factorization (ALS) to serve dynamic web carousels.
- **Probabilistic CLV Forecasting:** Implement the BG/NBD (Beta-Geometric / Negative Binomial Distribution) and Gamma-Gamma model suite via lifetimes to forecast transaction counts and monetary value dynamically.